

ALEJANDRO MAYORAL

+52 8119991144 | alexmayoral8@gmail.com | Engineering Portfolio

Biomedical Engineer with experience in medical device manufacturing, test engineering, manufacturing quality, equipment maintenance, incoming quality control, fixture design, and technical documentation within an ISO 13485-certified environment. Skilled in supporting manufacturing operations, developing test protocols and work instructions, performing preventive and corrective maintenance, improving production processes, and implementing quality controls.

RELEVANT EXPERIENCE

Bioana — Product Development Engineering Intern

Monterrey, Mexico | Dec 2024 - Dec 2025

- Supported engineering and manufacturing activities in an ISO 13485-certified medical device company, ensuring compliance with quality and documentation requirements.
- Performed preventive and corrective maintenance on equipment used for medical device manufacturing.
- Developed test protocols, test reports, acceptance criteria, and work instructions to support manufacturing quality and operator execution.
- Designed and fabricated test and manufacturing fixtures to support verification activities, improve repeatability, and reduce process variability.
- Designed and implemented incoming quality control tests to evaluate components, materials, and manufacturing readiness.
- Executed experiments and process evaluations in a cleanroom manufacturing environment, reducing process time by approximately 50% in an intraocular lens production step.
- Participated in failure analysis, PFMEA activities, and risk mitigation initiatives to improve manufacturing robustness.
- Collaborated with engineering, quality, and operations teams to improve manufacturing processes and define additional verification activities.

Research Assistant - Characterization of the coupling response between the BCI system-user through the analysis of EEG signals

Jan 2025 - June 2025

- Contributed to the project by conducting literature review, modifying code, processing data and analyzing results to identify patterns in EEG signals.
- Develop skills in signal processing, programming in MATLAB and Python, as well as interpretation of neurophysiological data.

EDUCATION AND COURSES

Universidad de Monterrey

Monterrey, Mexico | Aug 2021 - May 2026

B.S. in biomedical engineering

Universidad Miguel Hernández

Elche, Spain | Jan 2026 - May 2026

Academic Exchange - Rehabilitation Technologies Research Project

- Designed and validated a rehabilitation assessment interface integrating a robotic system and EMG acquisition.
- Conducted structured testing and performance evaluation with users.
- Processed and analyzed test data using MATLAB and Python.
- Documented procedures, methodologies, and technical findings.
- Presented the project at an academic symposium, receiving an “Accésit” recognition for its contribution.

COURSES AND CERTIFICATIONS

- Course in medical devices: “Bachelor Master Classes por Boston Scientific”.
- Course in Animal and Human Cell Culture: Applications in Regenerative Medicine (Universidad Complutense de Madrid).
- Certification as Microsoft Office Specialist: Excel
- Course in Data Analysis with Python by IBM

SKILLS

- Test Protocol & Report Development
- Incoming Quality Control (IQC)
- Root Cause Analysis
- PFMEA
- Fixture Design
- MATLAB, Python, Inventor, Fusion 360
- Microsoft Office: excel, word, power point
- Data analysis